

Adapting to climate change: Long-term impact of wind resource changes on China's power system resilience

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Abstract

Modern society’s reliance on power systems is at risk from the escalating effects of wind-related climate change. Yet, failure to identify the intricate relationship between wind-related climate risks and power systems could lead to serious short- and long-term issues, including partial or complete blackouts. Here, we develop a comprehensive framework to assess China’s power system resilience across various climate change scenarios, enabling a holistic evaluation of the repercussions induced by wind-related climate change. Our findings indicate that China’s current wind projects and planning strategies could be jeopardized by wind-related climate change, with up to a 12% decline in regional wind power availability. Moreover, our results underscore a pronounced vulnerability of power system resilience amidst the rigors of hastened climate change, unveiling a potential amplification of resilience deterioration, even approaching fourfold by 2060 under the most severe scenario, relative to the 2020 benchmark. This work advocates for strategic financial deployment within the power sector aimed at climate adaptation, enhancing power system resilience to avert profound losses from long-term, wind-influenced climatic fluctuations.

1 Introduction

Climate change is one of the most urgent and serious issues faced by all humanity today [1]. This multifaceted phenomenon has precipitated an escalated occurrence and severity of extreme events [2], posing a huge threat to human communities and infrastructure [3]. Besides, evidence suggests that there has been accelerated melting of glaciers [4] and an associated rise in sea levels [5], with considerable impacts on coastal and low-lying regions [6]. Such shifts contribute to ecosystem degradation [7], potentially leading to food and water shortages [8, 9] that further exacerbate challenges for human sustainability [10]. In light of these concerns, the Intergovernmental Panel on Climate Change (IPCC) [11] has exhorted national governments to reduce greenhouse gas emissions [12], especially carbon dioxide emissions [13], by promoting the energy transition as a pivotal strategy for combating climate change [14].

In China, the power sector constitutes the principal contributor to the nation’s carbon emissions [15], responsible for approximately 48% of the nation’s total emissions as reported by the International Energy Agency [16]. To facilitate a low-carbon energy transition [17], multiple decarbonization strategies have been initiated within this sector [18]. Most salient among these is the shift from a predominant reliance on coal-fired thermal power generation [19] to the incorporation of renewable energy sources (RES) [20, 21]. However, the large-scale integration of RES also brings substantial complexities to power system operations [22]. Renewable energy generation is inherently uncertain [23] owing to variable natural conditions of weather factors [24], which poses great challenges to maintaining real-time equilibrium between electricity

supply and demand during power system operations [25]. Wind energy, as a significant component of RES [26], accounted for approximately 14.2% of the total installed capacity within China’s power infrastructure as of 2022 [27, 28], also holding substantial prominence in the nation’s strategic development initiatives to fulfill energy transition and carbon neutrality objectives [29]. Yet, wind energy exhibits unique characteristics [30]. Its pronounced intermittency and variability [31, 32] can induce both high-probability, low-impact (HPLI) and low-probability, high-impact (LPHI) events [2], such as fluctuating wind power outputs [33] and extreme events like hurricanes [34], respectively, which can compromise the stability of power system operations [35], e.g., decreasing power system performance and even leading to blackouts [36].

Resilience is a critical metric for assessing the impact of HPLI and LPHI events on power system operations [37]. It refers to the ability to recover to normal operation as soon as possible after anticipating, withstanding external shocks and disturbances, and adapting to better cope with future events [38]. Assessing power system resilience helps in rational policy formulation and capital allocation in the power sector [39]. The IPCC also highlighted the necessity for adaptive strategies, including strengthening the resilience of infrastructure and ecosystems, to better mitigate the inescapable consequences of climate change [40]. Empirical evidence, however, indicates that climate change will not only alter future wind resource availability [41] but also intensify the frequency and severity of wind-related extreme events [42, 43]. Such climatic-induced shifts in wind resources hold the potential to further affect power system resilience, while they are often overlooked in extant literature [44]. In addition, when assessing the impact of climate change on power system resilience, relevant studies, which are notably limited, often adopted an energy-supply-centric perspective [2, 45–47] (that is, assessing resilience by considering changes in future renewable energy supply potential), while ignoring the physical characteristics of the power system operations, such as grid congestion [48–50] and generating unit ramp rates [51, 52]. These evaluations can be regarded as energy resilience rather than power system resilience. Considering operational integrity is a basis for assessing power system resilience [38]. Such an omission risks inaccurate resilience assessments, potentially leading to misguided policy-making and investment decisions with significant economic ramifications and even major operational accidents like blackouts.

In this article, we formulated a comprehensive framework to scrutinize the influence of wind-related HPLI and LPHI events on power system resilience, examining the possible impacts of long-term wind-related climate change on China’s power system resilience. It is also the first work to integrate wind-related climate change considerations with a comprehensive analysis of complex power system operations, marking a novel approach in the evaluation of power system resilience. First, we deployed simulations to project future variations in China’s wind resources under different shared socio-economic pathways (SSPs) [53] by combining multiple global climate models (GCMs), and then analyzed the evolving potential for wind power generation in China based on these SSP scenarios. Next, we constructed China’s province-level power system operation model with intrinsic physical characteristics, which were often overlooked by extant literature yet crucial for operational integrity, and its

resilience assessment framework to simulate future scenarios, allowing us to quantitatively appraise the consequences of wind-related HPLI and LPHI events on system resilience. Conclusively, we provided specific analyses and recommendations aimed at enhancing the long-term resilience of China’s power system, enabling its adaptation to future wind-related climate change.

2 Results

2.1 Wind resource potential assessment

We utilized the output from five GCMs (CanESM5 [54], CanESM5-1 [54], EC-EARTH5-VEG [55], MRI-ESM2-0 [56], and MIROC6 [57]) to project the climatic parameter of wind speed (m/s) from the period from 2020 to 2060 under four SSPs (SSP2-4.5, SSP3-7.0, SSP4-6.0, and SSP5-8.5, refer to Supplementary Extended Data Table 1). The ensemble mean of these five models was computed to represent a robust estimate of wind speed under different SSPs. To provide a detailed assessment of China’s wind resource potential, we averaged wind speeds within individual provinces’ grid cells, yielding province-level wind resource estimates. The temporal evolution of wind resource potential in China from 2020 to 2060 is presented in Fig. 1.

Our results reveal that wind resource potential decreases noticeably across several Chinese regions, particularly the western territories, as we move from SSP2-4.5 to SSP5-8.5. Yet, within each SSP scenario, the wind potential delineated across the Chinese map exhibits heterogeneous trajectories. Under the SSP2-4.5 scenario, most regions demonstrate a positive trend, suggestive of augmented wind resource potential, especially in China’s central regions. Only a few areas, possibly in the north and west, show a slight decrease. In contrast, under SSP3-7.0, an overall negative trend appears, especially pronounced in the southern territories, with few enhancements observed in the northern and northwestern areas. Different from the previous two scenarios, the SSP4-6.0 scenario, characterized by intensified climate change, exhibits a gamut of variations in wind resource potentials, from a decrement of 14% to an increment of 17%. Most regions, particularly the southeast, show an increasing trend, while the remaining regions, mainly in the southwest, reveal downtrends. This dynamic complexity is amplified under SSP5-8.5, where climate change impacts are further intensified. Wind potential variations fluctuate between -17% and 18%, with the southeast registering substantial gains contrasted by significant reductions in the southwest. In summary, our projections suggest that owing to the effects of climate change, China’s wind resource potential from 2020 to 2060 will display considerable heterogeneity across the considered SSP scenarios, with the SSP5-8.5 scenario heralding the most disparate outcomes.

Fig. 2 traces the evolution of wind resource potential and its conversion into wind power generation across Chinese provinces. This breakdown reveals the disparities in wind resource potential and wind power generation capability at a provincial level. Intriguingly, the invariability of wind speed in Xinjiang throughout all SSPs and its consistent power generation capability provide stable opportunities for wind energy

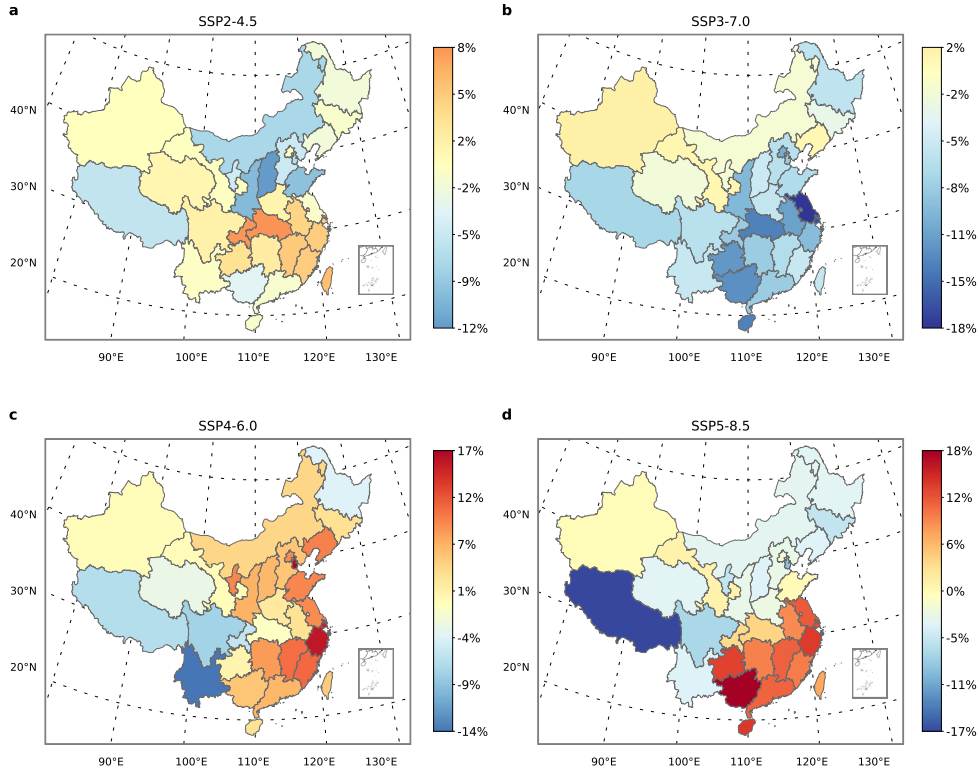


Fig. 1: Temporal evolution of wind resource potential in China from 2020 to 2060. a, SSP2-4.5. b, SSP3-7.0. c, SSP4-6.0. d, SSP5-8.5. The geographical maps delineate variations in wind resource potential across Chinese provinces, as indicated by color gradients. These potentials are inferred from the multi-model mean wind speeds, over GCMs of CanESM5, CanESM5-1, EC-EARTH5-VEG, MRI-ESM2-0, and MIROC6. Within each province, the wind resource potential is ascertained by averaging grid cell data.

investments. Conversely, regions such as Tibet, Hainan, and Shanghai display significant fluctuations in both wind speed and wind power generation capability, signaling potential challenges or opportunities in harnessing wind energy resources.

Fig. 2 panel b illuminates the tangible implications of these wind resource potentials on power production. Notably, under the SSP4-6.0 and SSP5-8.5 scenarios, provinces such as Fujian, Zhejiang, Guangxi, and Guangdong stand poised for an intensified capability in wind power generation, with the latter scenario indicating a particularly substantial upswing. This trend is consistent with these provinces' coastal locations and inherent capabilities in wind energy harnessing. Conversely, certain

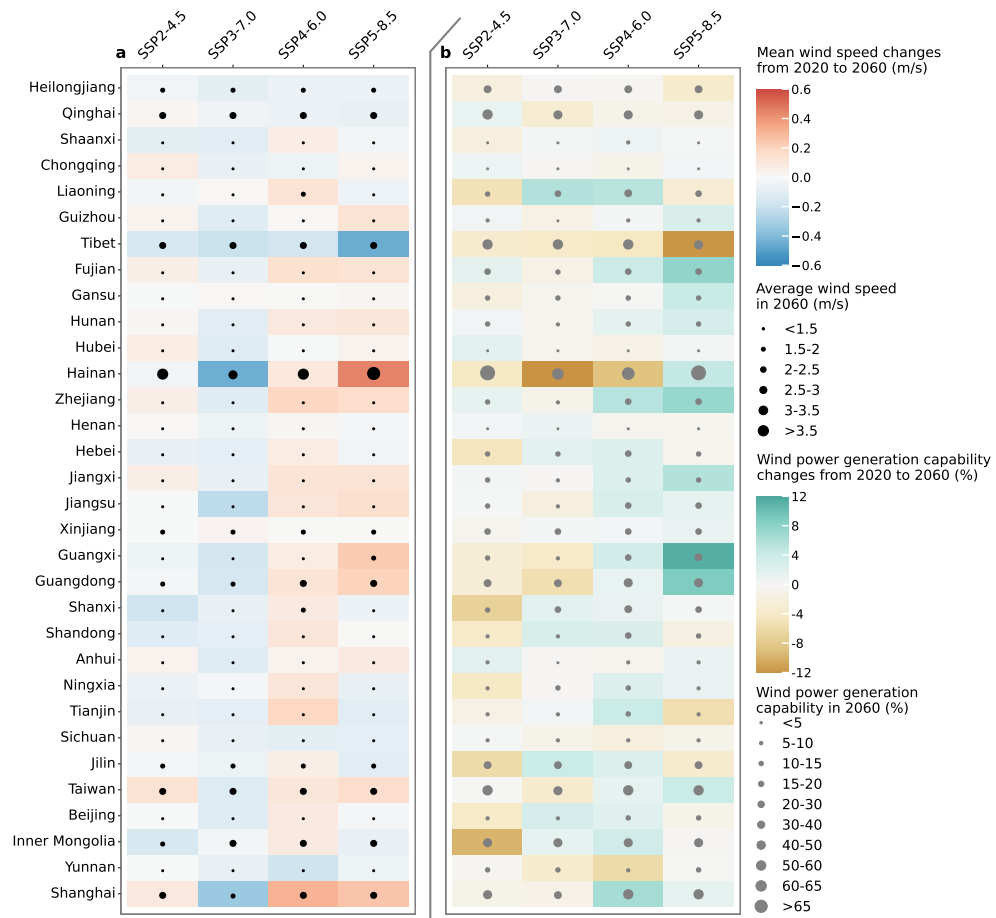


Fig. 2: Provincial wind resource potential and generation capability. **a**, wind speed, the color gradient elucidates the variance in wind speeds between 2020 and 2060, while the size of the black dot is proportionate to the annual mean wind speed specific to each province in 2060. **b**, wind power generation capability, the color scale portrays the deviation in wind power generation capabilities from 2020 to 2060, and the size of the gray dot correlates with the annual average wind power generation capacity specific to each province in 2060. The metric for wind power generation capability (expressed in %) is derived by normalizing the power yielded at the observed wind speed against the rated capacity, averaged over the annual period (see the “Methods” section).

165 provinces such as Jilin and Inner Mongolia (within the SSP2-4.5 scenario), Hainan and
 166 Guangdong (within the SSP3-7.0 scenario), Tibet and Hainan (within the SSP4-6.0
 167 scenario), and Heilongjiang and Tibet (within the SSP5-8.5 scenario) are predicted to

experience considerable downtrends by 2060. Despite their historically sizeable wind power generation capabilities, these provinces could witness a decline of up to 12% at most, potentially highlighting the challenges amplified by intensified climate change impacts. Particularly in Tibet, which consistently exhibits a differential yet discernible decline in wind power generation capability across all SSP scenarios, future investments in its wind energy infrastructure should be carefully assessed considering this projected trajectory.

Moreover, Tibet holds a significant position in China’s wind energy strategic planning, especially in aiding the realization of the nation’s carbon neutrality, energy transition, and emission reduction goals. Yet, China’s current wind energy planning in Tibet has no consideration of climate change impacts. This oversight of climate change’s long-term repercussions could precipitate substantial investment wastage and potentially imperil the safety and stability of the future power system. Hence, as wind energy projects in Tibet continue to advance, it becomes imperative to assiduously consider the impacts of climate change to safeguard the efficacy of the investments and the sustainability of the future power system. Employing scientific simulation and analysis to devise a more rational and anticipatory wind energy blueprint that addresses potential climate change impacts is instrumental in ensuring the triumphant execution of energy strategies in Tibet and across the nation.

The unique situations in provinces such as Qinghai and Jilin also warrant attention. Despite experiencing only minor changes in wind speed, their capability to convert wind resources into actual power generation varies significantly. This variation underscores the cubic relationship between wind speed and wind power, emphasizing the importance of baseline wind resource availability and geographical factors. Interestingly, provinces like Hubei and Anhui under the SSP3-7.0 scenario, project minimal changes in wind power generation capabilities, even when confronted with substantial wind speed fluctuations. This implies that their inherent wind resource potential remains relatively limited, making them less sensitive to future wind speed changes. As a result, wind energy investments in these provinces and the resultant power generation capability appear to be largely unaffected by potential future wind resource fluctuations.

In conclusion, while wind speeds serve as a foundational metric for estimating wind resource availability [58], the conversion of these resources into practical wind power remains contingent upon complex climate change effects. Our results highlight the heterogeneous wind resource potential across China’s provinces, each displaying unique trajectories in their wind power generation capabilities up to 2060. Policy-makers, therefore, must demonstrate astute discernment of these complex variations in wind resource potential and their subsequent power generation capabilities across Chinese provinces, against the backdrop of diverse climate change scenarios. Armed with such insights, it becomes possible to formulate strategic wind power development plans that optimize capital allocation, by avoiding overspending in provinces with increasing potential, and preventing underfunding in those with declining capabilities. This proactive approach is indispensable for staving off long-term disequilibria between power supply and demand, which could otherwise erode the resilience of the power system.

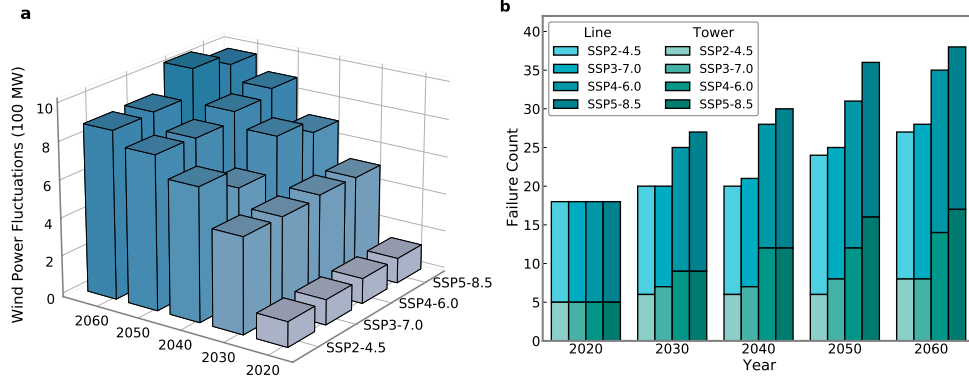


Fig. 3: Temporal trajectory of wind-related events across varied SSP scenarios. **a**, HPLI event, denoting the degree of fluctuation in wind power generation, quantified via the annual mean standard deviation of provincial wind power generation. **b**, LPHI event, illustrated through the annual failure frequency of transmission lines and towers. Both HPLI and LPHI events in 2020 provide a benchmark for comparison.

2.2 Power system resilience assessment

We conducted an assessment of China’s power system resilience for the years 2020, 2030, 2040, 2050, and 2060, under the SSP2-4.5, SSP3-7.0, SSP4-6.0, and SSP5-8.5 scenarios. Projections for future electrical power demand were based on China’s development plan (see the “Methods” section for more details). We determined the expected increase in capacities across various types of power generation from a pathway toward China’s carbon neutrality [59] (see Supplementary Extended Data Fig. 1). Resilience serves as a critical evaluative metric in analyzing the power system’s operational performance in response to HPLI and LPHI events. For this study, we systematically simulated wind-related HPLI and LPHI events based on the wind resource conditions of the aforementioned SSP scenarios. These were subsequently incorporated into power system operations to thoroughly evaluate China’s power system resilience (see the “Methods” section).

Fig. 3 maps out the temporal dynamics of wind-related HPLI and LPHI events across the various SSP scenarios. In subfigure **a**, there emerges a discernible increase in the volatility of wind power generation from 2020 to 2060. The onset of 2020 starts with relatively lower fluctuations, implying more stable wind power outputs. Yet, as the timeline advances to 2060, an amplification in fluctuations is observed. This trend, potentially indicative of a broader climatic difference induced by climate change, highlights the growing unpredictability of wind power generation. Upon examining the SSP variants, it becomes apparent that the SSP2-4.5 and SSP3-7.0 scenarios consistently display a modest fluctuation across the time frame. In contrast, the SSP4-6.0 and SSP5-8.5 scenarios, while initiating with modest fluctuations, exhibit a notable

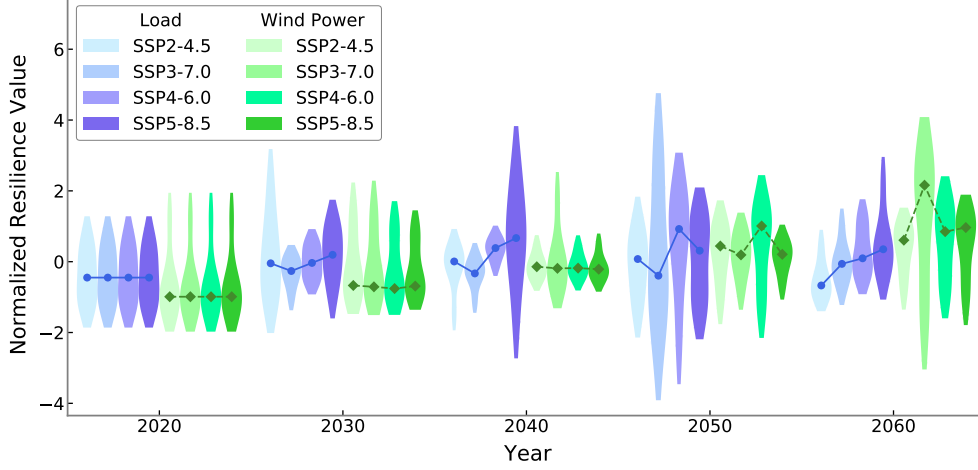


Fig. 4: Trajectory of power system resilience affected by wind-related HPLI and LPHI events across varied SSP scenarios. Each violin plot encapsulates the standard normalized resilience values of six resilience metrics $\mathcal{M}_1$, $\mathcal{M}_2$, $\mathcal{M}_3$, $\mathcal{M}_4$, $\mathcal{M}_5$, and $\mathcal{M}_6$ (see the “Methods” section). The median values of these normalized values, spanning the four SSP scenarios for a designated year, are interconnected by solid (for load) or dotted (for wind power) lines. A lower normalized value indicates a more resilient power system against these wind-related events.

intensification, surpassing the other scenarios by 2060. Such findings imply a prospective elevated variability in wind power output under the SSP4-6.0 and SSP5-8.5 projections, underscoring that intensified climate change could amplify wind power uncertainties.

In Fig. 3 b, an escalating trend of annual failure frequency of transmission lines and towers (LPHI events) is evident. From 2020 to 2030, the trend remains relatively stable, but a steep rise in failure instances is apparent post-2030, especially after 2040. By 2060, the failure count reaches its peak, underscoring the high-impact nature of these infrequent but severe LPHI events over time. Different SSP scenarios portray varied patterns. For transmission lines, SSP5-8.5 consistently reports higher failure counts, highlighting its vulnerability. For transmission towers, SSP5-8.5 also shows an alarming spike, especially in 2050 and 2060, signaling pronounced risks to transmission tower infrastructure under this scenario. These increased failure rates and risks are emblematic of the challenges that accelerated climate change scenarios could pose, necessitating a re-evaluation of power system resilience and adaptability strategies.

Fig. 4 presents the temporal evolution of power system resilience affected by wind-related HPLI and LPHI events across varied SSP scenarios. The illustrated violin plots integrate six standard normalized resilience metrics ($\mathcal{M}_1$ to $\mathcal{M}_6$, detailed in the “Methods” section), where an elevated normalized value indicates diminished power system resilience against wind events. From 2020 to 2060, there emerges a discernible decline in China’s power system resilience under the influence of climate change.

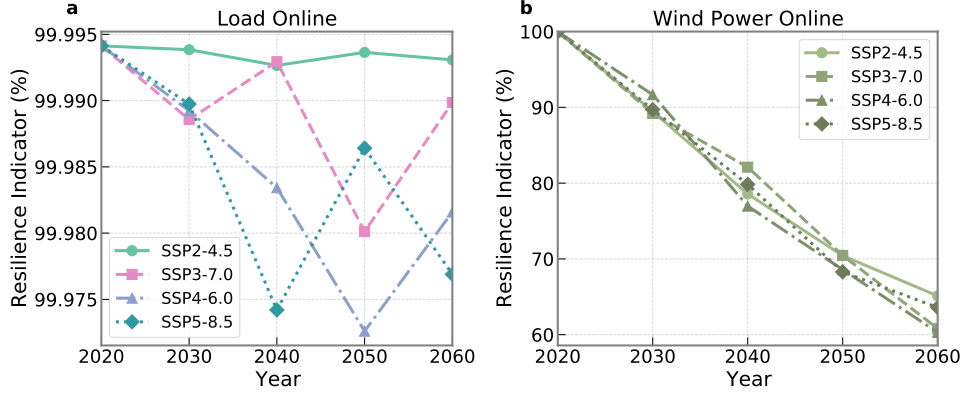


Fig. 5: Temporal evolution of power system resilience affected by wind-related HPLI and LPHI events across varied SSP scenarios. a, load online trajectory. b, wind power online trajectory. For each SSP scenario, the solid or dotted line signifies the annual mean value pertaining to either the load online or wind power online.

A close examination of the SSP trajectories underscores the pronounced ramifications of heightened climate change scenarios. The SSP4-6.0 and SSP5-8.5 scenarios, in particular, display evident vulnerabilities, as illustrated by their expansive value distributions and elevated medians. When compared with other scenarios, these two delineate a heightened sensitivity to climatic adversities. Even the SSP2-4.5 scenario, which assumes moderate climate change, is not immune. While all scenarios suggest eroding resilience over time, the SSP4-6.0 and SSP5-8.5 scenarios are markedly more disturbed. These findings emphasize the imperative of forward-thinking strategies to navigate anticipated severe climate scenarios, thereby safeguarding the sustainability and resilience of China's power system.

This finding is further corroborated in Fig. 5, which presents the annual load and wind power online rates, serving as key resilience indicators. In subfigure a, the load online rate seems to maintain a high resilience level across all scenarios. Yet, even minor dips below a 100% load online rate could allude to power outages or blackouts, which are significant accidents stemming from inadequate resilience against wind-related disturbances. Specifically, the resilience related to load experiences noticeable declines under climate change, particularly in SSP3-7.0, SSP4-6.0, and SSP5-8.5. Quantitative assessments reveal a slight 1.2-fold reduction in resilience for SSP2-4.5 by 2060 relative to the 2020 benchmark, where the load offline rate deteriorated from an initial 0.0058% to 0.0069%. The most stringent climate scenario, SSP5-8.5, witnesses a resilience downturn by 3.9-fold (from 0.0058% to 0.0231%), suggesting a potential quadrupling in blackout occurrences in parts of China due to compromised resilience.

Fig 5 b underscores a palpable decline in the resilience associated with wind power across all scenarios spanning 2020 to 2060. In contrast to the load online rates, the decrement in the wind power online rates is manifestly prominent. However, their

implications diverge. Unlike the load online rate not achieving 100%, which is indicative of significant blackout events, wind power rates falling short of 100% imply potential wind power generation curtailment. Even though the power system might lack the resilience to accommodate such surpluses, these surpluses do not necessarily lead to operational disruptions. Yet, such curtailments could impinge upon the power system’s economic viability. Harnessing this otherwise curtailed wind energy could pave the way for reduced coal-fired power operations, curtailing fuel costs and carbon emissions, thereby advancing the pursuit of carbon neutrality. Climate change exhibits the same impacts on wind power curtailment, wherein accelerated climate change scenarios (SSP4-6.0 and SSP5-8.5) can have more severe influences than others. Augmenting climatic alterations, particularly within the SSP4-6.0 and SSP5-8.5 trajectories, may exacerbate these wind power curtailment implications, impeding China’s power sector’s efforts toward carbon neutrality.

With the intensification of climate change, wind-driven HPLI and LPHI events are expected to amplify in frequency and magnitude. Such climatic anomalies, coupled with changing wind resource potentials, could significantly impact China’s power system operations. Given the surging incorporation of wind energy to meet carbon neutrality goals, these wind-related climatic shifts could pose a substantial challenge to China’s power system resilience.

3 Discussion and Suggestion

As climate change accelerates, China’s power system resilience may experience considerable challenges. Our simulation results show that wind-related climate change might instigate sizable alterations in wind resources throughout China. Such shifts might, on the one hand, pose significant concerns for wind power investments in regions experiencing dwindling wind potentials, such as Tibet. Investors might grapple with challenges in securing their returns on investment due to underperformed wind power outputs throughout the asset’s life cycle. On the other hand, China’s power system resilience might also be affected. Unanticipated alterations in wind resource potential, absent rigorous climate change assessment, might engender regional discrepancies in wind power output across China, disrupting the equilibrium of long-term power supply and demand and, consequently, impinging upon power system operations. While certain climate change scenarios may theorize that regions witnessing a dip in wind power production could be offset by counterparts experiencing gains, the inherent physical characteristics of the power system, such as transmission capacity, could impede the redistribution of wind power from high-output regions to those facing shortfalls. Given the physical constraints on the ramp rate of other generating units, the power system may often resort to load-shedding to sustain operations. Such predicaments, however, may become more prevalent under pronounced climate change scenarios, wherein the power system lacks insufficient resilience to navigate the rigors of exacerbated wind power fluctuations (HPLI events).

Wind-related extremities (LPHI events) may also challenge China’s power system resilience. Our findings indicate that with the progression of climate change, the frequency of wind-related LPHI events could surge significantly. This uptick can be

325 attributed to the pronounced disturbances in the atmospheric steady-state equilibrium
326 due to exacerbated climate change. While this may result in a redistribution of wind
327 resources, it can concurrently amplify wind speed volatilities, culminating in prevalent
328 strong wind and LPHI events. The amplified wind strength concomitantly elevates the
329 likelihood of damage to various power system infrastructures, such as transmission
330 lines and towers, thereby altering China's power system topological structure. Con-
331 sequently, owing to inadequate power system resilience, some regions may be unable
332 to meet local power demands post-event, necessitating the implementation of load-
333 shedding operations to sustain power system functionality. On the other hand, should
334 intensified wind speeds surpass the cut-out threshold, wind turbines would cease oper-
335 ation to preserve mechanical integrity and prevent damage, resulting in an abrupt
336 transition of wind power generation from maximal to null output. If the power sys-
337 tem lacks sufficient resilience to handle such sudden alterations in power generation,
338 power failures in certain areas may become more frequent. Thus, in the context of
339 climate change, China's power system resilience may exhibit a downward trajectory,
340 attributed to intensified wind events.

341 These signs indicate that China's power system resilience needs a pressing enhance-
342 ment as a proactive adaptation to climate change, which can be navigated via several
343 directions. First, power system planning and expansion must meticulously integrate
344 climate change impacts. This necessitates policymakers to exhibit acute perceptiveness
345 in discerning the prospective trajectory of wind resource alterations amidst China's
346 climate change landscape. Informed strategic investments and construction initiatives,
347 derived from such insights, aim to forestall long-term power supply and demand imbal-
348 ances resultant from shifts in wind power generation capability, thereby enhancing
349 power system resilience against wind resource shifts. Second, the expeditious augmen-
350 tation of sufficient flexible resources is paramount. Anticipating that climate change
351 might invoke more pronounced wind speed variations and, given the cubic relationship
352 between wind speed and wind power, potentially engender greater intermittency and
353 variability in wind power generation, more pervasive short-term imbalances between
354 supply and demand may ensue. Flexible resources, such as battery energy storage
355 systems, can be regarded as an effective resilience enhancement measure to counter
356 such imbalances swiftly, mitigating the risks of instability in power system operations
357 induced by wind power fluctuations. Third, the hardening of critical power system
358 infrastructures is imperative. Accelerating climate change may heighten the frequency
359 and severity of wind-related extremities, such as hurricanes, posing tangible threats to
360 power system infrastructures. Strategies for hardening critical infrastructures (includ-
361 ing updating and hardening lines, strengthening poles and towers with guy wires,
362 and burying lines underground) are urgently required to diminish susceptibilities to
363 wind-related damages, thus enhancing power system resilience and climate adaptation.
364 Furthermore, the development of advanced demand-side management strategies, like
365 demand response, also deserves exploration. Such strategies can modulate users' power
366 demand in tandem with volatile power generation resources at critical junctures, par-
367 ticularly amidst future high renewable energy penetrations, thereby fortifying power
368 system resilience. Nonetheless, all resilience-hardening measures require substantial
369 financial budgets and allocations. To this end, China's power sector, in pursuit of

carbon neutrality, must meticulously evaluate the potential impact of climate change on power system resilience to formulate judicious strategies, circumventing ineffective fiscal allocations.

Methods

Weather data sets from climate models

We used GCMs to simulate weather data from 2020 to 2060. As individual GCMs are predicated on divergent assumptions and deploy varying computational algorithms to model intricate Earth systems processes, using an ensemble approach serves to mitigate uncertainty and improve the predictive skill for extreme events [60]. Here, we synthesized (via arithmetic mean) outputs from five GCMs, namely CanESM5 [54], CanESM5-1 [54], EC-EARTH5-VEG [55], MRI-ESM2-0 [56], and MIROC6 [57], to project weather data under SSP2-4.5, SSP3-7.0, SSP4-6.0, and SSP5-8.5 scenarios, offering a comprehensive view of plausible climatic futures. In this regard, the monthly temperature and wind speed from the synthesized mean output were used in subsequent experiments.

Simulation of wind-related HPLI and LPHI events

We employed the monthly wind speed data synthesized from the aforementioned five GCMs to simulate wind-related HPLI and LPHI events. To transition from monthly to continuous hourly wind speed datasets, the Weibull distribution [46] was employed through learning historical continuous wind speed data. The associated probability density function for this distribution is expressed as,

$$f(w; \lambda, k) = \frac{k}{\lambda} \left(\frac{w}{\lambda} \right)^{k-1} e^{-(w/\lambda)^k} \quad (1)$$

where w denotes wind speed, while $\lambda > 0$ and $k > 0$ represent the scale and shape parameters, respectively, obtained by the maximum likelihood estimation via fitting historical continuous wind speed data. These parameters allow the Weibull distribution to model a variety of distributional forms, especially in wind speed [61, 62], due to its ability to capture the variability and randomness inherent in wind behavior. Then, the continuous hourly wind speed can be simulated by generating a series of uniformly distributed random variables, such that $r \sim U(0, 1)$, into the inverse function of the cumulative distribution function of the fitted Weibull distribution, as follows,

$$w = \lambda (-\ln(1 - r))^{\frac{1}{k}} \quad (2)$$

In this study, a scale factor was computed by contrasting future and historical monthly average wind speeds. This scale was then applied to historical hourly wind speed data for adjustment, mimicking the behavior of future wind speed. After that, the consequent adjusted data were modeled using the Weibull distribution, yielding the simulated hourly wind speed values across diverse SSP scenarios. This approach offers an effective strategy for continuous hourly wind speed simulation, contributing to the modeling of wind-related HPLI and LPHI events.

406 For HPLI events, the essence is the uncertain fluctuations in wind power outputs.
 407 Utilizing simulated hourly wind speeds, we computed the power generation metrics for
 408 wind farms. Fundamental to this calculation are certain wind speed thresholds char-
 409 acterized by their specific physical implications. The cut-in wind speed delineates the
 410 minimal wind speed required for initiating power generation. Within the wind speed
 411 range extending from the cut-in to the rated speed, a cubic relationship exists between
 412 wind speed and the resultant power output. Beyond the rated speed and up to the cut-
 413 out speed, power generation plateaus at a maximum constant value. At wind speeds
 414 exceeding the cut-out threshold, the turbine ceases operation to safeguard mechanical
 415 integrity and avoid damage. The functional relationship between wind speed w and
 416 power output $P^{\text{Wind}}(w)$ is formalized as follows (also illustrated in Supplementary
 417 Extended Data Fig. 2 (a)),

$$P^{\text{Wind}}(w) = \begin{cases} 0, & \text{if } w < w_{\text{cut-in}} \\ \beta(w^3 - w_{\text{cut-in}}^3), & \text{if } w_{\text{cut-in}} \leq w < w_{\text{rated}} \\ P_{\text{rated}}, & \text{if } w_{\text{rated}} \leq w < w_{\text{cut-out}} \\ 0, & \text{if } w \geq w_{\text{cut-out}} \end{cases} \quad (3)$$

418 where $P^{\text{Wind}}(w)$ denotes the power generated at a given wind speed w . The terms
 419 $w_{\text{cut-in}}$, w_{rated} , $w_{\text{cut-out}}$ signify the cut-in, rated, and cut-out wind speeds, respec-
 420 tively. The coefficient β encapsulates the effect of blade angle on wind energy capture
 421 efficiency. P_{rated} specifies the maximum sustainable power output.

422 For LPHI events, fragility curves are deployed to articulate the failure probability of
 423 a component as a function of the weather parameter [38], *i.e.*, wind speed, as exhibited
 424 in Supplementary Extended Data Fig. 2 (b). The component includes transmission
 425 towers and transmission lines [63]. By mapping the simulated wind speed profile at
 426 each simulation step to these fragility curves, the wind- and time-dependent failure
 427 probability of transmission towers and lines in a power system can be obtained, as
 428 follows,

$$\mathcal{P}^{\text{Failure}}(w) = \begin{cases} 0, & \text{if } w < w_{\text{critical}} \\ (1 + e^{-(aw+b)})^{-1}, & \text{if } w_{\text{critical}} \leq w < w_{\text{collapse}} \\ 1, & \text{if } w \geq w_{\text{collapse}} \end{cases} \quad (4)$$

429 where $\mathcal{P}^{\text{Failure}}(w)$ denotes the failure probability function with wind speed w . The
 430 threshold w_{critical} is the value of the weather parameter at which the failure probability
 431 picks up, whilst w_{collapse} indicates the juncture wherein the component is certain to
 432 fail. Parameters a and b are the steepness and bias of the logistic function, respectively.
 433 It is imperative to note that for transmission towers and lines, the values of w_{critical} ,
 434 w_{collapse} , a and, b are intrinsically tied to the specific component.

435 In simulations, the LPHI event occurs when $\mathcal{P}^{\text{Failure}}(w) > r$ (that is, specific trans-
 436 mission towers and lines will trip or fail), wherein r is defined as a uniformly distributed
 437 random variable such that $r \sim U(0, 1)$. The recovery time for both transmission
 438 infrastructures is subject to wind-related damages.

Electrical power demand

We projected China’s electrical power demand trajectory through to 2060 based on the Research Report on China’s “14th Five-Year” Power Development Plan [64] and the Report on China’s Long-term Low-carbon Development Strategy and Pathway [65]. The power demand growth rates for the milestones of 2030, 2040, and 2050 were ensured from these two reports [59]. For the 2060 projection, we adopted a mean growth rate, derived by averaging the aforementioned power demand growth rates. In addition to that, we considered the influence of temperature [47] on power demand (i.e., cooling and heating) for each SSP, derived from the spatial outcome from the ensemble of five GCMs. For each province, its temperature was represented by the mean of gridded temperatures within this province. Specifically, for temperatures below 16°C, a 1°C decrement in the temperature correlates with a 0.98% increase in electricity consumption for heating [66]. Conversely, for temperatures exceeding 28°C, a 1°C ascent in the temperature is associated with a 0.63% increment in electricity requisitioned for cooling [67]. Then, we applied the ascertained scaling factor to harmonize with province-specific hourly power demands in 2020. Such a refined hourly demand projection was integrated into dynamic power system operations, facilitating a comprehensive assessment of future power system resilience.

Modeling of power system operations

We developed an integrated framework for the optimal regulation of dynamic activities within power systems. This integrated approach incorporates three principal models: a dynamic direct current (DC) optimal power flow (OPF) model [68], a load-shedding model [69], and a wind power curtailment model [70].

In the dynamic DC-OPF model, the principal objective function, delineated by Eq. (5), aims to minimize the total cost of power generation. An essential constraint Eq. (6) is to ensure power balance (that is, the power injected in a bus meets the power generated in this bus minus its bus load). The model also needs to account for the maximum capacity of each transmission line, as shown in Eq. (7). Particularly important in the dynamic model, each generator has upper and lower bounds that should consider the generating unit ramp rates, as shown in Eq. (8). The DC-OPF model uses the linear power flow as Eq. (9) to relate the phase angles of the voltage at each bus to the power flows on each line.

$$obj. : \quad \min \sum_{i \in \Omega_{\text{Gen}}} C_i(P_{i,t}) \quad (5)$$

$$s.t. : \quad \sum_{(i,j) \in \Omega_{\text{Line}}} P_{i,j,t} = P_{i,t}^{\text{Bus}} - L_{i,t}^{\text{Bus}}, \quad \forall i \in \Omega_{\text{Bus}} \quad (6)$$

$$|P_{i,j,t}| \leq P_{i,j}^{\text{max}}, \quad \forall i \in \Omega_{\text{Line}} \quad (7)$$

$$\underline{P}_{i,t}^{\text{Gen}} \leq P_{i,t}^{\text{Gen}} \leq \bar{P}_{i,t}^{\text{Gen}}, \forall i \in \Omega_{\text{Gen}} \quad (8)$$

$$P_{i,j,t} = \frac{\theta_{i,t} - \theta_{j,t}}{X_{i,j}}, \forall (i,j) \in \Omega_{\text{Line}} \quad (9)$$

where the time-specific subscript t signifies the pertinent hour. Ω_{Gen} , Ω_{Line} , Ω_{Bus} represent sets of generators, transmission lines, and buses, respectively. The function $C_i(P_{i,t})$ characterizes the cost function affiliated with generator i , while $P_{i,t}$ is the generator's power output at time t . Power generation and load at bus i are represented as $P_{i,t}^{\text{Bus}}$ and $L_{i,t}^{\text{Bus}}$, respectively. The line power flow between buses i and j is denoted as $P_{i,j,t}$, which is determined by the voltage phase angles $\theta_{i,t}$ and $\theta_{j,t}$ at buses i and j under this line's reactance $X_{i,j}$, and cannot exceed its capacity $P_{i,j}^{\text{max}}$. The dynamic boundaries for power output from generator i are described by $\underline{P}_{i,t}^{\text{Gen}}$ and $\bar{P}_{i,t}^{\text{Gen}}$, respectively, which should take into account the generating ramp rates of conventional units and the real-time generation capabilities of renewable units.

In the load-shedding model, systematic load reduction is executed in response to insufficient supply or compromised network infrastructure. The goal of the objective function Eq. (10) seeks to minimize the overall negative impact of such shedding activities (that is, the associated cost of load shedding needs to be as little as possible). A distinctive constraint between the load-shedding model and the OPF model is that the electricity supply may fall short of demand, as captured in Eqs. (11) and (12).

$$\text{obj. :} \quad \min \sum_{i \in \Omega_{\text{Gen}}} C_i(P_{i,t}) + \sum_{i \in \Omega_{\text{Bus}}} C^{\text{Shed}}(\Delta L_{i,t}^{\text{Bus}}) \quad (10)$$

$$\text{s.t. :} \quad (7), (8), (9)$$

$$\sum_{(i,j) \in \Omega_{\text{Line}}} P_{i,j,t} = P_{i,t}^{\text{Bus}} - (L_{i,t}^{\text{Bus}} - \Delta L_{i,t}^{\text{Bus}}), \forall i \in \Omega_{\text{Bus}} \quad (11)$$

$$0 \leq \Delta L_{i,t}^{\text{Bus}} \leq L_{i,t}^{\text{Bus}}, \forall i \in \Omega_{\text{Bus}} \quad (12)$$

where the function $C^{\text{Shed}}(\Delta L_{i,t}^{\text{Bus}})$ indicates the associated cost of load shed $\Delta L_{i,t}^{\text{Bus}}$ at bus i .

The wind power curtailment model addresses the challenge of surplus wind energy. The objective function is similar to the OPF model. However, unlike conventional units, this model permits the power generated by wind units to operate below their typical power output levels, governed by the constraints stipulated in Eqs. (13) and (14).

$$\text{obj. :} \quad (5)$$

s.t. : (6), (7), (9)

$$P_{i,t}^{\text{Gen}} \leq P_{i,t}^{\text{Gen}} \leq \bar{P}_{i,t}^{\text{Gen}}, \forall i \in \Omega_{\text{Gen}} - \Omega_{\text{Wind}} \quad (13)$$

$$0 \leq P_{i,t}^{\text{Gen}} \leq P_{i,t}^{\text{Wind}}, \forall i \in \Omega_{\text{Wind}} \quad (14)$$

where Ω_{Wind} is a set of wind units and $P_{i,t}^{\text{Wind}}$ is the wind power without curtailment.

This comprehensive framework provides a multi-faceted approach to dynamic power system operations, contributing a set of models and constraints designed for optimized system performance and resilience.

Resilience assessment

We constructed an evaluation framework to assess power system resilience when its operations encountered wind-related HPLI and LPHI events. The objects assessed in power system resilience included system-level load shedding and wind power curtailment, which undergo three phases: disturbance progress (Phase I), post-disturbance degraded (Phase II), and restorative phase (Phase III) (illustrated in Supplementary Extended Data Fig. 3). Six resilience metrics were considered in the evaluation framework, including how fast ($\mathcal{M}_1$, unit: MW/h) and how low ($\mathcal{M}_2$, unit: MW) resilience drops in Phase I, how extensive ($\mathcal{M}_3$, unit: h) is the post-event degraded state in Phase II, how promptly ($\mathcal{M}_4$, unit: MW/h) the power system recovers to its pre-event resilience state in Phase III, the total impact ($\mathcal{M}_5$, unit: MW·h) in all Phases, and the resilience drop frequency ($\mathcal{M}_6$, unit: times) in a yearly horizon. To synthesize these six metrics, their specific standard normalization was implemented to eliminate the influences of units. Within this standard normalization, a greater value denotes a worse power system resilience and vice versa.

Moreover, another critical indicator was devised to quantify power system resilience, offering a numerical assessment of the online status of both load and wind power. This indicator is mathematically represented as,

$$S_t^{\text{Online}} = \frac{S_t - \Delta S_t}{S_t} \quad (15)$$

where S_t^{Online} symbolizes the online rate of either load or wind power. S_t indicates the system-level demand or wind power generation, and ΔS_t represents the magnitude of load shedding or wind power curtailment. The aggregate annual resilience measure, pertaining to either load or wind power retention, is derived by averaging these values over the temporal span of a year. A higher online rate is indicative of superior power system resilience, and the opposite holds true for a lower online rate. It is noteworthy to emphasize that, given the rarity of load-shedding operations (only occurred in a few moments through a whole year and typically consequent to significant disruptions and failures), a marginal decline in the load online rate (averaged by all moments) indicates a considerable deterioration in power system resilience.

526 Uncertainty analyses

527 We assessed power system resilience by considering a series of potential uncertainties.
528 The parameterizations within these simulations were subjected to stochastic variations
529 relative to their original values, specifically: (1) the growth rate of power demand
530 across provinces from 2020 to 2060 ($\pm 1\%$); (2) fluctuations in hourly power demand
531 ($\pm 5\%$); (3) fluctuations in hourly wind power generation for HPLI events ($\pm 2\%$); (4)
532 continuous hourly wind speed simulation via the Weibull distribution ($\pm 2\%$); (5) the
533 LPHI events through re-sampling of random uniform distribution to determine the
534 potential failures in transmission towers and lines; (6) fluctuations in the recovery time
535 of damaged transmission lines and towers induced by LPHI events (± 1 hour).

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543 Author contributions

544 J.R., Z.X., and J.Z. conceived and designed the research. J.R., G.L. Z.X., and J.Z.
545 formulated the theoretical model. X.M. and Y.Z. conducted the code implements and
546 carried out the simulations. J.R., X.M., Y.Z., G.L., X.S., H.W., H.X., M.L., P.G., J.L.,
547 and W.W. conducted the data analysis and led the writing of the paper. J.R. wrote
548 the first draft of the paper. J.R., X.M., and Y.Z. contributed to the data collection and
549 figure drawing. J.R., G.L., Z.X., and J.Z. conducted the policy analysis. All authors
550 contributed to the discussions on the framework and the editing of this article.

551 Competing interests

552 The authors declare no competing interests.

553 Data availability

554 The wind speed and temperature simulated by GCMs for different SSPs used in
555 this study are available from the Coupled Model Intercomparison Project Phase
556 6 (CMIP6) in the World Climate Research Programme (WCRP) [[https://esgf-
557 node.llnl.gov/search/cmip6/](https://esgf-node.llnl.gov/search/cmip6/)]. The historical wind speed and temperature data
558 used in this study are available from the ERA5-Land hourly data in the
559 Climate Data Store [[https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-
560 era5-land?tab=form](https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-land?tab=form)]. Data used in this study are publicly available on GitHub
561 [<https://github.com/ruanjiaqi01/China-Power-System-Resilience>].

Code availability

Codes used in MATLAB and Python for this study are available from the authors upon reasonable request.

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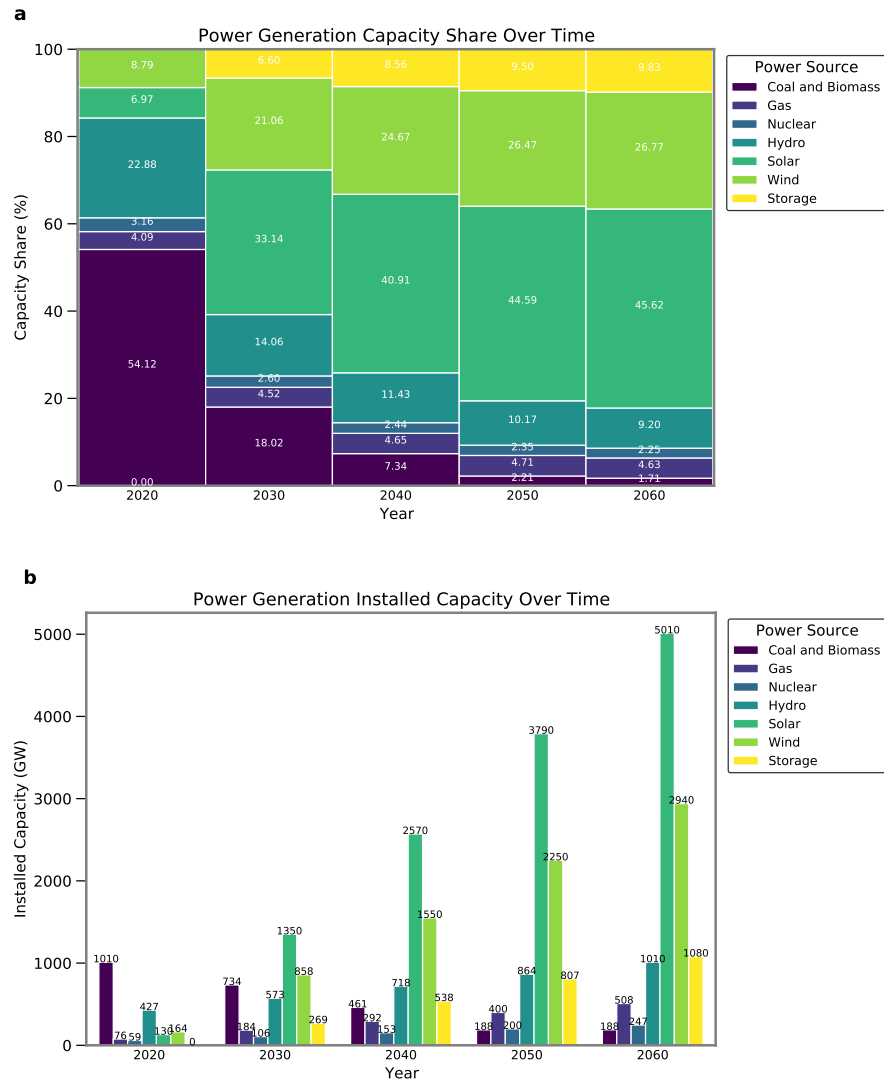
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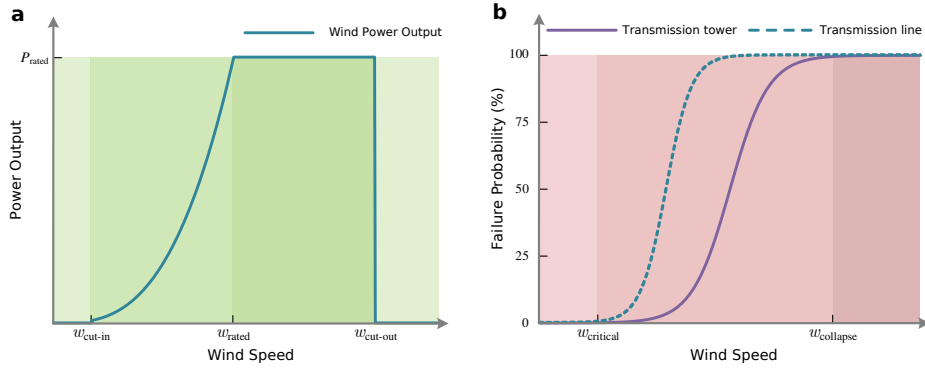
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Extended Data Table 1 Brief introduction of SSP scenarios selected in simulations.

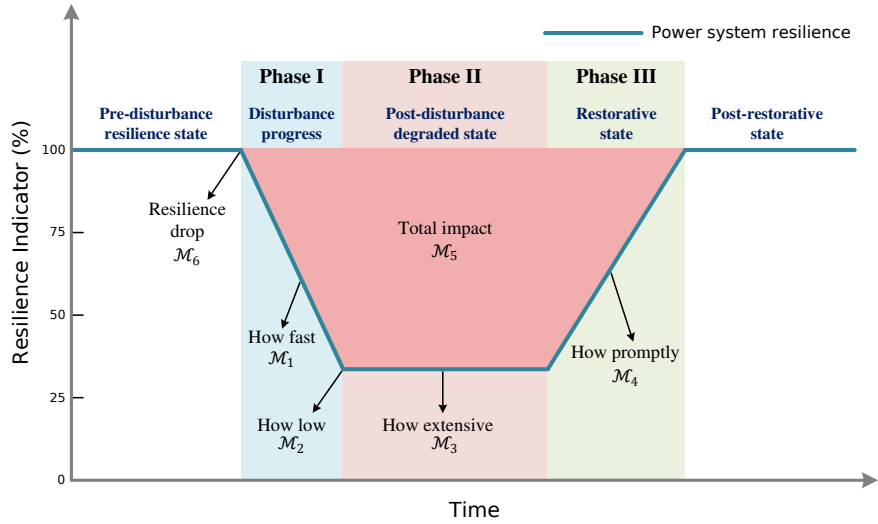
Scenario	Brief Introduction
SSP2-4.5	Middle of the Road: A world evolving without significant societal shifts, leading to a radiative forcing of 4.5 W/m ² by 2100. This scenario presents a moderate challenge to both climate mitigation and adaptation.
SSP3-7.0	Regional Rivalry: A fragmented world marked by regional focuses and resurgent nationalism, resulting in a radiative forcing of 7.0 W/m ² by 2100. It poses a high challenge to adaptation and a medium challenge to mitigation.
SSP4-6.0	Inequality: A world characterized by increasing disparities, where a minority enjoys prosperity while the majority struggles, leading to a radiative forcing of 6.0 W/m ² by 2100. This scenario presents a high challenge to mitigation and a medium challenge to adaptation.
SSP5-8.5	Fossil-fueled Development: A world pursuing rapid growth powered predominantly by fossil fuels, resulting in a high radiative forcing of 8.5 W/m ² by 2100. It poses a significant challenge to mitigation efforts but a lesser challenge to adaptation.



Extended Data Fig. 1 Temporal evolution of power generation capacity from 2020 to 2060. a, Power generation capacity share over time. b, Power generation installed capacity over time.



Extended Data Fig. 2 Illustration of the relationship of wind speed and HPLI and LPHI events. **a**, the function between wind speed and wind power output. **b**, the fragile curves under different wind speeds, in which the y-axis indicates the failure probability for the transmission tower and line.



Extended Data Fig. 3 Illustration of power system resilience. Three phases are indicated by different colors. Six resilience metrics are considered, including how fast ($\mathcal{M}_1$) and how low ($\mathcal{M}_2$) resilience drops in Phase I, how extensive ($\mathcal{M}_3$) is the post-event degraded state in Phase II, how promptly ($\mathcal{M}_4$) the power system recovers to its pre-event resilience state in Phase III, the total impact ($\mathcal{M}_5$) in all Phases, and the resilience drop frequency ($\mathcal{M}_6$) in a yearly horizon.